

Quantum Transport in Topological Insulator Thin Films

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Introduction

Topological insulators (TIs) host metallic surface states protected by time-reversal symmetry, while the bulk remains insulating. These states are promising for spintronics and low-dissipation electronics.

- Bulk band gap: 0.3 eV
- Dirac surface states with spin-momentum locking
- Robust against non-magnetic disorder

Theory

The low-energy surface Hamiltonian is

$$H = \hbar v_F (\sigma_x k_y - \sigma_y k_x), \quad (1)$$

where $v_F \approx 5 \times 10^5 \text{ m s}^{-1}$ and σ_i are Pauli matrices. The resulting Dirac cone is shown in Fig. 1.

Methods

Thin films (5–20 nm) were grown by MBE on $\text{Al}_2\text{O}_3(0001)$. Transport measurements were performed in a dilution refrigerator down to 20 mK.

Results

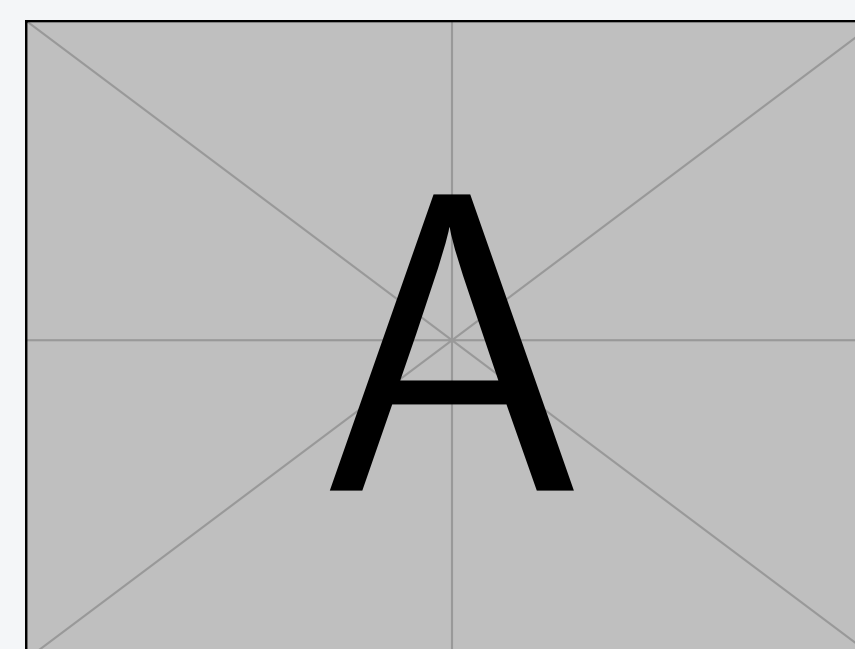


Figure 1: Longitudinal conductivity vs. temperature.

Discussion

The data are consistent with weak antilocalization from strong spin-orbit coupling. Deviations below 50 mK suggest bulk leakage, which we attribute to inadvertent doping.

Conclusions

- High-quality TI films grown by MBE.
- Surface-dominated transport above 50 mK.
- Future work: gate-tuning the Fermi level.

References

- M. Z. Hasan and C. L. Kane, Rev. Mod. Phys. **82**, 3045 (2010).
- X.-L. Qi and S.-C. Zhang, Rev. Mod. Phys. **83**, 1057 (2011).